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### Bottle assembly

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#### Abstract

A bottle assembly includes a container, a collar assembly, and a vent insert. The collar assembly includes a collar and a nipple. The vent insert is configured for selective placement on an open top of the container. The vent insert, when placed on the open top of the container, defines a gap for allowing liquid in the liquid chamber of the container to flow towards the nipple and a vent passage having at least one vent opening in flow communication with the liquid chamber for allowing air to flow into the liquid chamber. The vent insert further defines an annular gap forming a portion of the vent passage. The bottle assembly is selectively operable in two different configurations, a first configuration in which the vent insert is placed on the open top of the container and a second configuration in which the vent insert is omitted.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/230,516 filed Apr. 14, 2021, which is a divisional of U.S. patent application Ser. No. 16/287,565 filed Feb. 27, 2019 (now U.S. Pat. No. 11,007,122), which claims the benefit of U.S. Provisional Patent Application No. 62/637,804 filed Mar. 2, 2018, the disclosures of which are hereby incorporated by reference in their entirety.

### FIELD

(1) The field of this invention relates generally to bottle assemblies and more particularly to a bottle assembly including a container, a nipple, a collar for securing the nipple to the container, and a cap for covering the nipple when the container is not in use.

### BACKGROUND

(2) Bottle assemblies, such as infant or nursing bottle assemblies, typically have multiple components including a bottle, a nipple, a collar for securing the nipple to the bottle (the nipple and collar sometimes collectively defining a collar assembly), and a cap for covering the nipple when the bottle is not in use. The nipple typically has one or more openings for allowing liquid contained within the bottle to exit through the nipple and into an infant's mouth for consumption by the infant (or young child). During use, the infant places an end of the nipple in their mouth and sucks on the nipple to withdraw the liquid contained within the bottle.

(3) Typical bottle assemblies require two hands to prepare the bottle for use and may be difficult to assemble. For example, the preparer may have to hold the bottle with one hand and remove the cap with the other hand to uncover the nipple. In addition, a user may have to grip and twist a collar to remove or secure the collar assembly on the container. However, the collar may be uncomfortable

and/or difficult for the preparer to grip and twist. These issues may be exacerbated because the preparer may have limited use of their hands. For example, the preparer may be holding a distressed infant and be simultaneously attempting to prepare the bottle.

(4) At least some bottle assemblies include a removable vent assembly that can be positioned within the bottle. Some examples of vented bottle assemblies include those available from Handi-Craft Company under the trade name Dr. Brown's. Additional examples are disclosed in U.S. Pat. No. 5,779,071 issued Jul. 14, 1998, U.S. Pat. No. 7,828,165 issued Nov. 9, 2010, U.S. Pat. No. 8,113,365 issued Feb. 14, 2012, and U.S. Pat. No. 8,146,759 issued Apr. 3, 2012, the disclosures of all of which are herein incorporated by reference in their entirety. In these bottle assemblies, the vent assembly allows air to enter the bottle while the infant consumes the liquid through the nipple, thus alleviating or reducing the formation of a vacuum within the bottle during nursing. The vent assembly typically seats, at least in part, on the rim of the bottle and a collar assembly including a collar and nipple are together threadably secured down over the vent assembly to external threads on the neck of the bottle. The vent assembly may vent the bottle to the atmosphere via a vent in the vent assembly that is located proximate the threads of the collar assembly. However, airflow into the vent could be inhibited by the amount of space available between the vent assembly and the collar assembly.

(5) There is a need, therefore, for a bottle assembly that may be easily and quickly assembled even when a preparer has limited use of their hands. In addition, there is a need for a vent assembly that provides improved airflow into the bottle assembly during use and prevents leakage.

#### SUMMARY

(6) In one aspect, a bottle assembly includes a container having an open end, a closed end, a base portion, and a neck together defining a liquid chamber within the container. The neck has a rim defining the open end of the container. The container has a central axis. The bottle assembly also includes a vent assembly positionable at least in part on the rim of the container to permit venting of the container during use. The vent assembly includes at least one lateral vent extending other than parallel to the central axis of the container when the vent assembly is positioned on the rim of the container. The vent assembly includes a vent insert including the at least one lateral vent and a perimeter wall. The perimeter wall defines at least one vent opening and at least partially defines an annular gap extending circumferentially around the vent insert. The at least one vent opening is in flow communication with the at least one lateral vent and the annular gap such that air is allowed to flow from the annular gap into the lateral vent through the vent opening. The bottle assembly further includes a collar assembly generally defining a closure for the container. The collar assembly is configured for releasable engagement with the neck of the container over the open end thereof to retain the vent assembly together with the container. The annular gap is defined between the collar assembly and the vent assembly.

(7) In another aspect, a bottle assembly includes a container having an open end, a closed end, a base portion, and a neck together defining a liquid chamber within the container. The container has a central axis. The neck has a rim defining the open end of the container. The neck of the container includes external threads thereon. The bottle assembly also includes a collar assembly generally defining a closure for the container. The collar assembly is configured for releasable engagement with the neck of the container over the open end thereof. The collar assembly includes a collar and a nipple. The collar includes an outer surface, an internal surface, and internal threads for threaded engagement with the external threads of the neck of the container. The internal threads are disposed on the internal surface. The outer surface includes a plurality of elongate grooves extending along the central axis of the container when the collar is secured to the container. The elongate grooves are configured to provide a grip for a user to remove the collar assembly from the container.

(8) In a further aspect, a bottle assembly includes a container having an open end, a closed end, a base portion, and a neck together defining a liquid chamber within the container. The neck has a rim defining the open end of the container. The bottle assembly also includes a collar assembly

generally defining a closure for the container. The collar assembly is configured for releasable engagement with the neck of the container over the open end thereof. The collar assembly includes a collar and a nipple. The bottle assembly further includes a cap including a closed end, an open end, and a side wall extending between the closed end and the open end. The closed end, open end, and side wall together defining an interior and an exterior. The cap is configured for releasable engagement with the collar assembly such that the nipple is received within the interior of the cap. The side wall includes a curved surface defining a recess for removal of the cap from the collar assembly. The curved surface is concave relative to the exterior of the cap. The recess being closer to the cap closed end than to the cap open end.

(9) In yet a further aspect, a bottle assembly includes a container having an open end, a closed end, a base portion, and a neck together defining a liquid chamber within the container. The neck has a rim defining the open end of the container. The container has a central axis. The bottle assembly also includes a vent assembly positionable at least in part on the rim of the container to permit venting of the container during use. The vent assembly includes a vent insert including an inner portion and an outer portion. The outer portion is configured to at least partially rest on the rim of the container. The inner portion includes an internal vent tube and a lip. The lip is cylindrical and extends along the central axis. The lip includes a plurality of bulges extending circumferentially around the lip. The vent assembly also includes a receptacle portion including a top and a tube. The top is configured to receive the lip of the vent insert. The plurality of bulges are configured to engage the top of the receptacle portion to form a seal between the vent insert and the top.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a nursing bottle assembly including a cap.
- (2) FIG. 2 is a perspective view of the nursing bottle assembly of FIG. 1 with the cap omitted.
- (3) FIG. 3 is a cross-section of the bottle assembly of FIG. 2.
- (4) FIG. 4 is a perspective view of a nipple of the bottle assembly of FIG. 1.
- (5) FIG. 5 is a side view of the nipple of FIG. 4.
- (6) FIG. 6 is a bottom view of the nipple of FIG. 4.
- (7) FIG. 7 is a perspective view of a collar of the bottle assembly of FIG. 1.
- (8) FIG. 8 is a side view of the collar of FIG. 7.
- (9) FIG. 9 is bottom view of the collar of FIG. 7.
- (10) FIG. 10 is a cross-section of the collar illustrated in FIG. 8.
- (11) FIG. 11 is an enlarged cross-section of a portion of the bottle assembly of FIG. 1 with the vent assembly included.
- (12) FIG. 12 is an enlarged cross-section of a portion of the bottle assembly of FIG. 1 with the vent assembly omitted.
- (13) FIG. 13 is a perspective view of the cap of the bottle assembly of FIG. 1.
- (14) FIG. 14 is a cross-section of the cap of FIG. 13.
- (15) FIG. 15 is a bottom view of the cap of FIG. 13.
- (16) FIG. 16 is an exploded view of the bottle assembly of FIG. 1.
- (17) FIG. 17 is a perspective view of a vent insert of the bottle assembly of FIG. 1.
- (18) FIG. 18 is a side view of the vent insert of FIG. 17.
- (19) FIG. 19 is a top view of the vent insert of FIG. 17.
- (20) FIG. 20 is a cross-section of a portion of the vent insert illustrated in FIG. 18.
- (21) FIG. 21 is a perspective view of a vent tube of the bottle assembly shown in FIG. 1.
- (22) FIG. 22 is a cross-section of a portion of a vent assembly including the vent insert of FIG. 17 and the vent tube of FIG. 21.

(23) FIG. 23 is a cross-section of a portion of the bottle assembly of FIG. 1 including a collar assembly retaining the vent assembly shown in FIG. 22 on a container of the bottle assembly.

(24) Corresponding reference characters indicate corresponding parts throughout the several views of the drawings.

#### DETAILED DESCRIPTION OF THE DRAWINGS

(25) Referring now to the drawings and in particular to FIGS. 1-3, a bottle assembly, generally indicated at **100**, includes a container **102** such as a bottle and more particularly a nursing bottle in the illustrated embodiment, and a collar assembly, generally indicated at **104** for generally closing the bottle. The container **102** includes a liquid chamber **106** defined therein and adapted to hold a quantity of liquid for consumption by a user, such as a nursing infant. It is also understood that the container **102** may be configured other than as illustrated herein, and may be configured for use other than as a nursing bottle assembly, such as a sports bottle, a drink tumbler, a training cup, a commuter cup, etc. The container **102** may be made of any suitable material such as, without limitation, glass, polypropylene or other plastic, aluminum, or stainless steel. The container **102** can also be made in any desired color or colors, and may be transparent, translucent, or opaque.

(26) The container **102** has a closed bottom **108**, an open top **110**, and a generally cylindrical side wall **112** extending between the closed bottom **108** and the open top **110**. The cylindrical side wall **112** includes a base portion **114** and a neck **116** that is narrowed with respect to the base portion **114**. That is, the neck **116** has a smaller diameter than the base portion **114**, as best seen in FIG. 16. It is understood that in other embodiments, the neck **116** diameter may only be slightly smaller than the diameter of the base portion **114**, or even the same diameter as the base portion, without departing from the scope of this invention. The neck **116** includes an annular rim **118**, an externally threaded portion **120**, and a shoulder **122** defined at the transition between the neck **116** and the base portion **114**. The threaded portion **120** includes threads **124** for assembling the container **102** to the collar assembly **104** as described later herein.

(27) The collar assembly **104** of the bottle assembly **100** is adapted for removable attachment to the container **102** for selectively holding a nipple **130** on the container **102**. The illustrated collar assembly **104** includes the nipple **130** and a collar **132**. The nipple **130** and the collar **132** can each be made of any suitable material. In one embodiment, for example, the nipple **130** is made of a substantially pliable material such as at least one of a rubber material, a silicone material, and a latex material, and the collar **132** is made of polypropylene. The nipple **130** and the collar **132** can be made in any desired color or colors, and may be transparent, translucent, or opaque.

(28) With reference to FIGS. 4-6, in the illustrated embodiment, the nipple **130** includes a nipple portion **135** and a transversely extending flange **134**. The nipple flange **134** includes a bottom face **136** that extends from a generally circular outer edge **138** to a generally circular inner edge **140**. In the illustrated embodiment, a peripherally extending lip **142** projects up from the flange generally adjacent the circular outer edge **138**. In at least some alternative embodiments, the nipple **130** does not include a peripherally extending lip **142**. The nipple portion **135** extends up from the flange **134** generally adjacent the circular inner edge **140** thereof. As illustrated in FIGS. 4 and 5, the nipple portion **135** includes an annular external projection **144** that projects radially outward. The nipple portion **135** also includes an outlet end **146** including an aperture **148** for dispensing liquid to the user. It is contemplated, however, that the nipple **130** can have different shapes and sizes than those illustrated and described herein without departing from the present invention.

(29) As shown in FIGS. 7-12, the collar **132** has an annular top panel **150** and a depending side wall **152**. The top panel **150** includes an annular projection **156** that extends downward from the underside of the panel proximate an annular nub or radially inner edge margin **154** of the top panel **150**. The edge margin **154** and the annular projection **156** facilitate assembly of the nipple **130** on the collar **132**. To assemble the collar assembly **104**, the nipple **130** is pulled, nipple portion **135** first, up through the central opening in the top panel **150** of the collar **132** until the edge margin **154** is positioned below the annular external projection **144** of the nipple **130**, and the annular

projection **156** of the collar **132** is positioned radially inward of the peripheral lip **142** of the nipple **130**. It is understood, however, that the nipple **130** and collar **132** may be configured other than as illustrated and still otherwise configured for assembly together for further assembly onto the container **102**. It is also contemplated that in other embodiments the nipple **130** and collar **132** need not be capable of being held in assembly for conjoint assembly onto the container **102**.

(30) The side wall **152** of the collar **132** has an inner surface **162** with suitable internal threads **164** for threaded engagement with the external threads **124** of the neck **116** of the container **102** to releasably secure the collar and hence the collar assembly **104** on the container **102**. In some embodiments, the number of threads **164** on the inner surface **162** of the collar side wall **152** exceeds the number of external threads **124** on the neck **116** of the container. In one embodiment, for example, the collar **132** may have approximately twice as many internal threads **164** as the number of external threads **124** on the neck **116** of the container **102**.

(31) As best seen in FIGS. **8** and **10**, the side wall **152** of the collar **132** has an outer surface **163** opposite the inner surface **162**. The outer surface **163** includes a plurality of grooves **165**. The grooves **165** are elongate, i.e., have a length that is greater than their width, and extend along a central axis **166** of the container **102** (shown in FIG. **1**) when the collar **132** is secured to the container. Accordingly, the grooves **165** are substantially perpendicular to the internal threads **164**. The grooves **165** are equally spaced circumferentially about the collar **132**. In the illustrated embodiment, the collar **132** includes eight grooves **165**. In other embodiments, the collar **132** may include any suitable grooves **165** without departing from some aspects of the disclosure.

(32) The grooves **165** help a preparer to grasp and rotate the collar **132** to adjust the threaded engagement of the container **102** and the collar and secure or remove the collar from the container. Specifically, the grooves **165** are configured to receive portions of a hand or other object used to grip the collar **132**. The grooves **165** increase the contact area between the object and the collar **132** and resist slippage of the object relative to the collar. Accordingly, the grooves **165** may provide an improved grip in comparison to at least some known collars. In addition, the user may feel less discomfort when gripping the collar **132** in comparison to at least some known collars because the collar includes grooves **165** and does not include projections or other grip structures which may cause high stress points on the gripping object.

(33) In the illustrated embodiment, each groove **165** extends from a bottom of the side wall **152** to beyond the midpoint of the side wall. Each groove **165** includes a first end **169** at the bottom of the collar **132** and a second end **171** defined by the side wall **152** at a point between the bottom and top of the collar. The second end **171** is at least partially rounded. The first end **169** is open and defines a maximum width of each groove **165**. The width of each groove **165** decreases from the first end **169** to the second end **171**. Accordingly, the second end **171** defines a minimum width of each groove **165**. In other embodiments, the grooves **165** may be any shape that enable the grooves to function as described herein. In some embodiments, the grooves **165** may be different shapes.

(34) As seen in FIGS. **1** and **13-16**, the bottle assembly **100** also includes a cap **101** for covering the nipple **130** when the bottle assembly **100** is not in use. The cap **101** includes an open end **103**, a closed end **105**, and a side wall **107** extending between the open end **103** and the closed end **105**. The side wall **107** and the closed end **105** define an interior space **109** sized and shaped to receive the nipple **130** therein. In the illustrated embodiment, the cap **101** is a dome. Specifically, the side wall **107** generally curves along the extension between the open end **103** and the closed end **105** such that the open end **103** has a greater diameter than the closed end **105**. Accordingly, the side wall **107** has a generally convex curve when viewed from the exterior of the cap **101**. In other embodiments, the cap **101** may have any shape that enables the cap **101** to function as described.

(35) The cap **101** may be positioned on the collar assembly **104** such that at least a portion of the collar assembly extends through the open end **103** and into the interior space **109**. Specifically, the nipple **130** is arranged to extend into the interior space **109** when the cap **101** is positioned on the collar assembly **104**. A plurality of projections **111** extend downward from the center of the closed



end **105** to engage the nipple **130** and inhibit displacement of the nipple when the cap **101** covers the nipple. In addition, the side wall **107** of the cap **101** includes latches **113** extending along the open end **103** that engage the collar **132**. The latches **113** extend radially inward from an inner surface of the side wall **107** and are sized to extend into an annular groove **115** (shown in FIG. **10**) along the base of the annular top panel **150** of the collar **132**. Accordingly, when the cap **101** is positioned on the collar assembly **104**, the edge **117** of the side wall **107** rests on the top of the side wall **152** and the latches **113** engage the collar **132**.

(36) In the illustrated embodiment, the cap **101** and the collar **132** are configured for a releasable snap-fit engagement. Specifically, to secure the cap **101** to the collar assembly **104**, the preparer positions the cap **101** on the collar assembly **104** and presses in a direction toward the collar **132**. Due to the pressing force, the cap **101** deforms slightly and allows the latches **113** to engage the collar **132**. To remove the cap **101** from the collar assembly **104**, the preparer presses or pulls the cap **101** away from the collar assembly **104** and the latches **113** disengage from the collar **132**. In other embodiments, the cap **101** may be secured to the collar **132** in any manner that enables the bottle assembly **100** to function as described herein.

(37) The cap **101** further includes a recess **119** defined by the side wall **107** and arranged for a preparer to grip the cap **101** and position the cap **101** relative to the collar assembly **104**. In the illustrated embodiment, the side wall **107** includes a curved surface **121** defining the recess **119**. The curved surface **121** has a radius of curvature that is different from the radius of curvature of the rest of the side wall **107**. Specifically, the curved surface **121** curves inward from the side wall **107** such that the recess **119** is concave relative to the exterior of the cap **101**. The recess **119** is sized and shaped to receive a thumb or other finger. For example, in the illustrated embodiment, the curved surface **121** include a perimeter edge **123** that defines an oval shape. In addition, the recess **119** is spaced from the open end **103** a distance that corresponds to an average finger length and is closer to the closed end **105** than to the open **103**. The distance between the open end **103** and the recess **119** allows a finger of a hand to be positioned in the recess while the same hand grasps the collar **132** when the cap **101** is secured to the collar assembly **104**. As a result, the recess **119** enables a person to remove the cap **101** from the collar assembly **104** using one hand. For example, the person may grasp the collar **132** of the bottle assembly **100** with one hand and press on the cap **101** with a finger of the hand positioned in the recess **119** to remove the cap **101** with a single hand.

(38) Suitably, the recess **119** is shaped and positioned to utilize the leverage of a person's finger and allow for easier positioning of the cap **101** relative to the collar assembly **104**. For example, the perimeter of the recess **119** includes two parallel circumferentially extending portions **126** connected by two axially extending arcs **128**. The parallel portions define outer edges of a middle portion of the recess **119** that is substantially U-shaped. The two arcs define outer edges of side portions which are partial bowl shapes. Accordingly, the recess **119** has an overall stadium shape, i.e., a rectangle with semicircles at opposite ends. The recess **119** is shaped for a user to apply a force at substantially a center of the recess **119**. In addition, the center of the recess **119** is closer to the closed end **105** than the open end **103**. Accordingly, when a finger is positioned in the recess and a force applied to the recess **119**, a moment equal to the distance between the center of the recess **119** and a pivot point is generated at the center of the recess **119** that facilitates removal of the cap. In the illustrated embodiment, substantially the entire recess **119** is located above a midplane of the cap **101**.

(39) With reference to FIGS. **16-21**, the bottle assembly **100** also includes a vent assembly **180** to permit venting of the bottle assembly during use. The vent assembly **180** includes a vent insert **182** and a receptacle portion **184**. The receptacle portion **184** is releasably attachable to the vent insert **182**. The receptacle portion **184** includes a top **186**, a middle portion **188**, and a vent tube **190**. The top **186** includes a generally cylindrical side wall **192** that defines a reservoir **194** therein. The vent tube **190** includes a generally cylindrical side wall **196** that has a smaller diameter than the side wall **192** of the top **186**. The side wall **196** of the vent tube **190** defines a passage **198** that is in

fluid communication with the reservoir **194**. The middle portion **188** includes a tapered side wall **200** that extends between the vent tube side wall **196** and the top side wall **192**. The vent tube **190** also includes an air outlet **202** at an end of the passage **198** proximate the closed bottom **108** of the container **102**.

(40) The vent insert **182** includes an inner portion **210** and an outer portion **212**. The inner portion **210** includes an internal vent tube **214**, a lip **216**, and a top wall **218** extending between and oriented orthogonal to the internal vent tube **214** and lip **216**. The internal vent tube **214** defines a passage **220** in flow communication with the reservoir **194**. The inner portion **210** of the vent insert **182** further includes a lateral vent **234** extending upward from the top wall **218** to define a channel **236** within the vent insert **182**. The outer portion **212** of the vent insert **182** includes a perimeter wall **240**. A gap **250** defined between the perimeter wall **240** and the top wall **218** enables liquid in the liquid chamber **106** to flow towards the nipple **130**. A plurality of ribs **251** extend across the gap **250** and connect the perimeter wall **240** to the top wall **218**. The lateral vent **234** extends to the perimeter wall **240** which has at least one vent opening **249** that allows the channel **236** to extend through the perimeter wall **240**. As best in FIG. 20, in the illustrated embodiment, the lateral vent **234** extends across the entire diameter of the vent insert **182** and is in flow communication with two vent openings **249** defined by the perimeter wall **240** on opposite ends of the lateral vent **234**.

(41) The perimeter wall **240** includes an inner edge **242**, an outer edge **244**, a top **246**, and a bottom **248**. The perimeter wall **240** also defines an annular gap **252** extending circumferentially around the vent insert **182** and in flow communication with the channel **236** via the vent openings **249**. Specifically, a curved surface **254** extends radially inward from the outer edge **244** towards the inner edge **242** to define the annular gap **252**. The curved surface **254** is concave relative to the outer edge **244** and is disposed between the top **246** and the bottom **248**. Accordingly, the thickness of the perimeter wall **240** defined between the outer edge **244** and the inner edge **242** is greater than the thickness of the perimeter wall **240** defined between the curved surface **254** and the inner edge **242**. Accordingly, the annular gap **252** provides increased space between the vent insert **182** and the collar assembly **104** to allow air flow **253** (shown in FIG. 23) into the channel **236** from the atmosphere through the collar assembly.

(42) With reference to FIG. 22, to assemble the vent insert **182** to the receptacle portion **184**, the lip **216** is inserted into the top **186** of the receptacle portion **184** such that the side wall **192** contacts the lip **216**. The lip **216** is configured to engage the side wall **192** of the receptacle portion **184**. For example, the lip **216** includes a plurality of annular ribs or bulges **222** that engage the side wall **192**. In the illustrated embodiment, the lip **216** is cylindrical and extends along the central axis **166** of the bottle assembly **100** (shown in FIG. 3). The bulges **222** extend circumferentially around the lip **216**. The bulges **222** are spaced axially apart along the lip **216**. The bulges **222** are continuous and provide a seal between the lip **216** and the side wall **192**. Moreover, the seal between the lip **216** and the side wall **192** is improved because the lip **216** includes two or more bulges **222**. In addition, the bulges **222** extend radially outward from an outer surface of the lip **216** to engage the side wall **192** when the lip is inserted into the receptacle portion. In other embodiments, the vent insert **182** may engage the receptacle portion **184** in any manner that enables the vent assembly **180** to function as described herein. For example, the bulges **222** may extend from an inner surface of the lip **216** and the receptacle portion **184** may be inserted in the lip **216**. In some embodiments, the receptacle portion **184** may include bulges **222** that engage the vent insert **182**. Alternatively, the vent insert **182** and the receptacle portion **184** may be integrally formed with each other.

(43) The lip **216** includes an outer surface **224** and an inner surface **226** opposite the outer surface **224**. The inner surface **226** is substantially smooth and defines a cylindrical bore through the lip **216**. A thickness of the lip **216** is defined between the inner surface **226** and the outer surface **224**. Each bulge **222** has a triangular cross-section and includes two equal sloped sides extending outward from the lip **216** and connecting at a peak. Accordingly, the lip **216** has a maximum thickness at the peaks of the bulges **222**.

(44) The bulges **222** are positioned to engage a portion of the side wall **192** spaced from the edge of the side wall **192** when the vent insert **182** and the receptacle portion **184** are assembled. A lowermost one of the bulges **222** is positioned at an edge of the lip **216**. The upper bulge **222** is positioned end to end with the lower bulge **222** and is spaced from the edge of the lip **216** by the width of the lower bulge **222**. The crests or peaks of the bulges **222** are spaced apart by a distance equal to or less than the width of one bulge **222**. In the illustrated embodiment, the lip **216** includes two bulges **222** that cover a majority, i.e., greater than 50%, of the outer surface **224** of the lip **216** and the lip **216** is free of any additional bulges or sealing features.

(45) It is contemplated, however, that the components of the vent assembly **180** can have different shapes and sizes than those illustrated and described herein without departing from some aspects of this invention. Similar vent assemblies are known in the art, such as those used on the bottle assemblies disclosed in U.S. Pat. No. 5,779,071 issued Jul. 14, 1998, U.S. Pat. No. 7,828,165 issued Nov. 9, 2010, U.S. Pat. No. 8,113,365 issued Feb. 14, 2012, and U.S. Pat. No. 8,146,759 issued Apr. 3, 2012.

(46) As illustrated in FIGS. **11** and **12**, the collar side wall **152** and the container neck **116** are suitably sized relative to each other to permit operation of the bottle assembly **100** in two different configurations, a first configuration (FIG. **11**) in which the vent assembly **180** is included and a second configuration (FIG. **12**) in which the vent assembly is omitted. For example, in the illustrated embodiment, the length (or height in the orientation of the drawings herein) of the container neck **116** (e.g., from the rim **118** of the container **102** to the shoulder **122** where the neck widens outward to the base portion **114** of the container) is sufficient to accommodate the side wall **152** of the collar in the second configuration, i.e., when the vent assembly is omitted as illustrated in FIG. **12**.

(47) Accordingly, as seen in FIGS. **11** and **23**, in the first configuration, where the bottle assembly **100** includes the vent assembly **180**, the vent assembly **180** is inserted into the container **102** such that the bottom **248** of the perimeter wall **240** seats down against the rim **118** of the container. After the vent assembly **180** is inserted within the container **102**, the collar assembly **104** is attached to the container **102** by threadably engaging the internal collar threads **164** with the external threads **124** of the neck **116** and rotating the collar **132** to twist the collar down onto the container. As the collar **132** is tightened onto the container **102**, the bottom face **136** of the nipple **130** is urged against the top **246** of the perimeter wall **240** of the vent assembly **180** to seal the nipple (and hence the collar assembly **104**) against the vent assembly. Concurrently, the bottom **248** of the perimeter wall **240** of the vent assembly **180** is urged against the rim **118** of the container **102** to seal the vent assembly against the container.

(48) In this configuration, some of the lower threads **164** of the collar engage the threads **124** of the neck, while the uppermost collar threads oppose the perimeter wall **240** of the vent assembly. As illustrated in FIG. **23**, the collar side wall **152** is sufficiently long such that in the first configuration, the lower end of the collar side wall extends below the lowermost threads **124** of the neck **116** so that no external threads are visible when the collar assembly **104** is secured to the container **102**. In addition, the annular gap **252** is defined between the perimeter wall of the vent assembly **180** and the internal surface of the collar including the threads **164**. Accordingly, air is permitted to flow along the threads **124** of the collar **132** and through the annular gap **252**. The airflow is allowed to flow throughout the annular gap **252** and into the passage **220** through the vent openings **249**. As a result, the annular gap **252** provides increased space for airflow and is believed to provide smoother airflow into the vent assembly **180**.

(49) In the second configuration, illustrated in FIG. **12**, the vent assembly **180** is omitted from the bottle assembly **100**. When the collar assembly **104** is tightened down onto the container **102** in this configuration, the collar threads **164** engage the threads **124** of the neck **116** of the container and the collar **132** is rotated to twist the collar down onto the container until the nipple **130** is urged against the rim **118** of the container to seal the nipple directly against the container. In this

configuration, a lower set of the collar threads is disposed below the lowermost external thread **124** of the neck **116**. In this manner, the lower end of the collar **132** is disposed well below the lowermost external thread **124** and nearer to the shoulder **122** of the container. In addition, the nipple **130** is configured to vent the container **102** to the atmosphere in the second configuration with the vent assembly **180** omitted. Specifically, the nipple **130** includes at least one vent feature **256** that allows airflow into the container **102** when the nipple **130** is positioned on the rim **118** of the container **102**.

(50) Embodiments of a bottle assembly described herein provide advantages over at least some known bottle assemblies. For example, the described bottle assemblies are simpler and more convenient for a preparer to assemble and may be at least partially assembled when the preparer has limited use of their hands. In addition, the collar assembly and the cap of the bottles assemblies may reduce discomfort during assembly and during use of the bottle assembly because the collar assembly and the cap reduce pressure points on the preparers hand(s) and provide an increased grip for the preparer. In addition, in some embodiments, the bottle assemblies include a vent assembly that provides increased airflow into the container from the atmosphere in comparison to known vented bottles. In addition, the vent assembly may include a double seal that allows for simpler assembly and reduces leakage in comparison to known vent assemblies.

(51) When introducing elements of the present invention or the various versions, embodiment(s) or aspects thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The use of terms indicating a particular orientation (e.g., “top”, “bottom”, “side”, etc.) is for convenience of description and does not require any particular orientation of the item described.

(52) As various changes could be made in the above without departing from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

## Claims

1. A bottle assembly comprising: a container defining a liquid chamber for holding a quantity of liquid, the container having a closed bottom, an open top, and a side wall extending between the closed bottom and the open top, the side wall including a base portion and a neck that is narrowed with respect to the base portion such that the neck has a smaller diameter than the base portion, the neck including an externally threaded portion and a shoulder defined at a transition between the neck and the base portion; a collar assembly including a collar and a nipple, the collar assembly being adapted for removable attachment to the container, the nipple including a nipple portion with an aperture and a transversely extending flange with a bottom face, the collar including a collar side wall and an opening, the nipple portion of the nipple extending through the opening in the collar, the collar side wall of the collar having an inner surface with internal threads for threaded engagement with the external threaded portion of the neck of the container to releasably secure the collar assembly on the container; and a vent insert configured for selective placement on the open top of the container, the vent insert, when placed on the open top of the container, defines a gap for allowing liquid in the liquid chamber of the container to flow past the vent insert towards the nipple and a vent passage having at least one vent opening in flow communication with the liquid chamber of the container for allowing air to flow into the liquid chamber, the vent insert further defining an annular gap, the annular gap forming a portion of the vent passage, wherein the bottle assembly is selectively operable in two different configurations, a first configuration in which the vent insert is placed on the open top of the container and a second configuration in which the vent insert is omitted, the annular gap defined by the vent insert being disposed above the open top of the container in the first configuration of the bottle assembly.

2. The bottle assembly set forth in claim 1, wherein the annular gap extends circumferentially around at least a portion of the vent insert.
3. The bottle assembly set forth in claim 1, wherein the annular gap is in direct flow communication with the at least one vent opening.
4. The bottle assembly set forth in claim 1, wherein the annular gap is defined between a perimeter wall of the vent insert and the inner surface of the collar side wall.
5. The bottle assembly set forth in claim 4, wherein the perimeter wall is a concaved wall.
6. The bottle assembly set forth in claim 5, wherein air is able to flow along the internal threads of the inner surface of the collar side wall and through the annular gap.
7. The bottle assembly set forth in claim 1, wherein the annular gap facilitates decreases turbulence of air flowing into the liquid chamber of the container.
8. The bottle assembly set forth in claim 1, wherein the nipple of the collar assembly is a vented nipple having a vent opening.
9. The bottle assembly set forth in claim 1, wherein the nipple of the collar assembly can be selectively removed from the collar of the collar assembly.
10. A bottle assembly comprising: a container defining a liquid chamber for holding a quantity of liquid, the container having a closed bottom, an open top, and a side wall extending between the closed bottom and the open top, the side wall including a base portion and a neck that is narrowed with respect to the base portion such that the neck has a smaller diameter than the base portion, the neck including an externally threaded portion and a shoulder defined at a transition between the neck and the base portion; a collar assembly including a collar and a nipple, the collar assembly being adapted for removable attachment to the container, the nipple including a nipple portion with an aperture and a transversely extending flange with a bottom face, the collar including a collar side wall and an opening, the nipple portion of the nipple extending through the opening in the collar, the collar side wall of the collar having an inner surface with internal threads for threaded engagement with the external threaded portion of the neck of the container to releasably secure the collar assembly on the container; and a vent insert configured for selective placement on the open top of the container, the vent insert, when placed on the open top of the container, defining a gap for allowing liquid in the liquid chamber to flow past the vent insert towards the nipple and a vent passage having at least one vent opening in flow communication with the liquid chamber of the container for allowing air to flow into the liquid chamber, the vent insert further defining an annular gap, the annular gap forming a portion of the vent passage, wherein the bottle assembly is selectively operable in two different configurations, a first configuration in which the vent insert is placed on the open top of the container and a second configuration in which the vent insert is omitted, the annular gap defined by the vent insert being disposed adjacent the collar assembly and above the open top of the container in the first configuration of the bottle assembly.
11. The bottle assembly set forth in claim 10, wherein the annular gap extends circumferentially around at least a portion of the vent insert.
12. The bottle assembly set forth in claim 10, wherein the annular gap is defined between a perimeter wall of the vent insert and the inner surface of the collar side wall.
13. The bottle assembly set forth in claim 12, wherein the perimeter wall is a concaved wall.
14. The bottle assembly set forth in claim 12, wherein air is able to flow along the internal threads of the inner surface of the collar side wall and through the annular gap.
15. The bottle assembly set forth in claim 10, wherein the annular gap facilitates decreases turbulence of air flowing into the liquid chamber of the container.
16. The bottle assembly set forth in claim 10, wherein the annular gap is formed, at least in part, by a curved surface of the vent insert.
17. The bottle assembly set forth in claim 10, wherein the nipple of the collar assembly can be removed from the collar of the collar assembly.
18. The bottle assembly set forth in claim 10, wherein the nipple of the collar assembly is a vented

nipple having a vent opening.

19. A bottle assembly comprising: a container defining a liquid chamber for holding a quantity of liquid, the container having a closed bottom, an open top, and a side wall extending between the closed bottom and the open top, the side wall including a base portion and a neck, the neck including an externally threaded portion; a collar assembly including a collar and a nipple, the collar assembly being adapted for removable attachment to the container, the nipple including a nipple portion with an aperture and a transversely extending flange with a bottom face, the collar including a collar side wall and an opening, the nipple portion of the nipple extending through the opening in the collar, the collar side wall of the collar having an inner surface with internal threads for threaded engagement with the external threaded portion of the neck of the container to releasably secure the collar assembly on the container; and a vent insert configured for selective placement on the open top of the container, the vent insert, when placed on the open top of the container, defining a gap for allowing liquid in the liquid chamber to flow past the vent insert towards the nipple and a vent passage having at least one vent opening in flow communication with the liquid chamber of the container for allowing air to flow into the liquid chamber, the vent insert further defining an annular gap, the annular gap forming a portion of the vent passage, wherein air flows along the internal threads of the inner surface of the collar side wall and into the annular gap defined between the vent insert and the collar assembly to flow through the vent passage of the vent insert into the liquid chamber, wherein the bottle assembly is selectively operable in two different configurations, a first configuration in which the vent insert is placed on the open top of the container and a second configuration in which the vent insert is omitted, the annular gap defined by the vent insert being disposed above the open top of the container in the first configuration of the bottle assembly.

20. The bottle assembly set forth in claim 19, wherein the annular gap extends circumferentially around at least a portion of the vent insert.

21. The bottle assembly set forth in claim 19, wherein the nipple of the collar assembly can be removed from the collar of the collar assembly.

22. The bottle assembly set forth in claim 19, wherein the nipple of the collar assembly is a vented nipple having a vent opening.

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